

# **MATH 100:**

# **Linear Algebra**

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# Topics Covered

- ▶ Vector Equations and Matrix Equations
- ▶  $Ax=b$  Equation
- ▶ Solution Sets of Linear Systems

## 1.3 Vector Equations

## Recap

- Augmented matrices lead us to consider rectangular arrays of numbers—matrices.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

## Definition of matrix

**Definition:** A **matrix** is a rectangular array of numbers.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

We say that  $A$  has  $m$  rows and  $n$  columns.

The **dimension** (or **size**) of  $A$  is  $m \times n$  ( $m$ -by- $n$ ).

The element  $a_{ij}$  is called the  $(i, j)$ th entry.

# Special Matrix Shapes

An  $m \times n$  matrix is a **square matrix of order  $n$**  if  $m = n$ .

A  $1 \times n$  matrix is a **row matrix** (or **row vector**):

$$(a_1 \quad a_2 \quad \cdots \quad a_n).$$

A  $m \times 1$  matrix is a **column matrix** (or column vector):

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix}.$$

**Abbreviation:** In **this** course we will often mention vectors. It is a column vector.

## Vectors in $\mathbb{R}^2$

A vector with two entries has the form

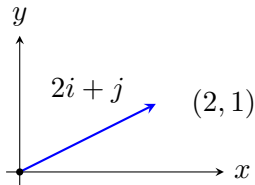
$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad u_1, u_2 \in \mathbb{R}.$$

## ordered 2-tuples

The set of all vectors with two entries is denoted by  $\mathbb{R}^2$  (read “R-two”).

The  $\mathbb{R}$  means the entries are real numbers, and the exponent <sup>2</sup> indicates that each vector contains two entries.

# Geometric Description of $\mathbb{R}^2$



point  $\iff$  vector  $\iff$  column matrix

$$(2,1) \qquad 2\mathbf{i} + \mathbf{j} \qquad \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Vector  $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$  is the point  $(x_1, x_2)$  in the plane.

$\mathbb{R}^2$  is the set of all points in the plane.



## Vectors in $\mathbb{R}^3$ and $\mathbb{R}^n$

Vectors in  $\mathbb{R}^3$  (vectors with 3 entries)

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (\text{an ordered 3-tuple}).$$

Vectors in  $\mathbb{R}^n$  (vectors with  $n$  entries)

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad (\text{an ordered } n\text{-tuple}).$$

written as  $\mathbf{u}$  or  $\vec{u}$ .

# The Zero Vector $\mathbf{0}$

The zero vector  $\mathbf{0}$

The vector whose entries are all zero.  $\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$

# The Zero Vector $\mathbf{0}$

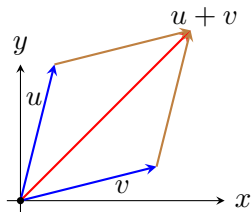
## The zero vector $\mathbf{0}$

The vector whose entries are all zero.  $\mathbf{0} =$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Well... appears useless right?

# Vector Arithmetic: Addition



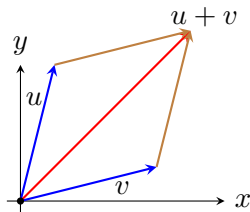
We can view each column as a vector.

$$u = \begin{bmatrix} 0.5 \\ 2 \end{bmatrix}$$

$$v = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

$$u + v = \begin{bmatrix} 2 + 0.5 \\ 2 + 0.5 \end{bmatrix}$$

# Vector Arithmetic: Addition



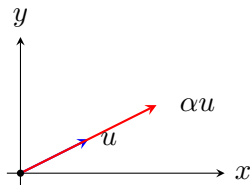
## Parallelogram Rule

If  $\mathbf{u}$  and  $\mathbf{v}$  are points in the  $\mathbb{R}^2$  plane:

We have three points represented by  $\mathbf{v}$ ,  $\mathbf{u}$ ,  $\mathbf{0}$  (origin).

Three points define a Parallelogram: it ends at  $\mathbf{u} + \mathbf{v}$ .

# Vector Arithmetic: Scalar Multiplication

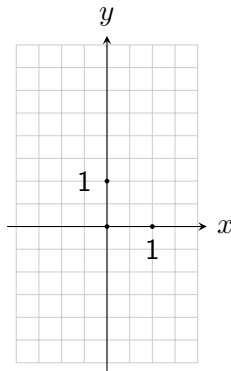


## Scalar Multiplication

For a scalar  $\alpha$  (this is a number), the vector  $\alpha \mathbf{u}$  stretches or shrinks  $\mathbf{u}$  correspondingly. Its direction is along the same line, but its length is scaled by  $|\alpha|$ . If  $\alpha < 0$ , its direction is reversed.

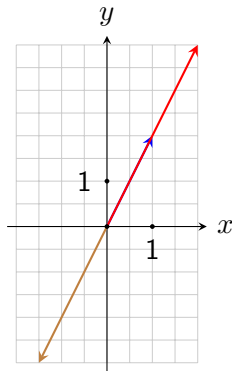
# Vector Arithmetic: Scalar Multiplication

Let  $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ . Express  $\mathbf{u}$ ,  $2\mathbf{u}$ , and  $-\frac{3}{2}\mathbf{u}$  on a graph.



# Vector Arithmetic: Scalar Multiplication

Let  $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ . Express  $\mathbf{u}$ ,  $2\mathbf{u}$ , and  $-\frac{3}{2}\mathbf{u}$  on a graph.





# Vector Addition: recap

$$u = \begin{bmatrix} 0.5 \\ 2 \end{bmatrix}$$

$$v = \begin{bmatrix} 2 \\ 0.5 \end{bmatrix}$$

$$u + v = \begin{bmatrix} 2 + 0.5 \\ 2 + 0.5 \end{bmatrix}$$

# Vector Addition: Geometry and Algebra

Arrows  $\longleftrightarrow$  Column Matrices  
 Operations on vectors  $\longleftrightarrow$  Matrix operations

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}.$$

# Scalar Multiplication: recap

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Express  $\mathbf{u}$ ,  $2\mathbf{u}$ , and  $-\frac{3}{2}\mathbf{u}$  on a graph:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad 2\mathbf{u} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad -\frac{3}{2}\mathbf{u} = \begin{bmatrix} -\frac{3}{2} \\ -3 \end{bmatrix}.$$

# Scalar Multiplication: Geometry and Algebra

|                       |                       |                   |
|-----------------------|-----------------------|-------------------|
| Arrows                | $\longleftrightarrow$ | Column Matrices   |
| Operations on vectors | $\longleftrightarrow$ | Matrix operations |
| Geometry              | $\dashrightarrow$     | Algebra           |

$$\alpha \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} \alpha u_1 \\ \alpha u_2 \end{pmatrix}.$$

# Vector Operations in $\mathbb{R}^2$

Let  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ .

- ▶  $\mathbf{u} = \mathbf{v}$  if and only if their corresponding entries are equal. (exactly equal!)
- ▶  $\mathbf{u} + \mathbf{v}$  is obtained by adding corresponding entries.
- ▶  $c\mathbf{u}$ , where  $c$  is a scalar, is obtained by multiplying each entry in  $\mathbf{u}$  by  $c$ .

Linear system is defined by solution sets.

Equality, vector addition, and scalar multiplication in  $\mathbb{R}^n$  are defined entry by entry, just as in  $\mathbb{R}^2$ .

# Eight Algebraic Properties

For all  $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$  and scalars  $\alpha, \beta$ :

$$\text{A1. } \mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$$

$$\text{A2. } (\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$$

$$\text{A3. } \mathbf{x} + \mathbf{0} = \mathbf{x}$$

$$\text{A4. } \mathbf{x} + (-\mathbf{x}) = \mathbf{0}$$

$$\text{A5. } \alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$$

$$\text{A6. } (\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$$

$$\text{A7. } (\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$$

$$\text{A8. } 1\mathbf{x} = \mathbf{x}$$

# Linear Combinations

Given  $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$  and scalars  $c_1, \dots, c_p$ , the vector

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

is called a **linear combination** of  $\mathbf{v}_1, \dots, \mathbf{v}_p$ , using weights  $c_1, \dots, c_p$ .

Examples include

$$\sqrt{3}\mathbf{v}_1 + \mathbf{v}_2, \quad \frac{1}{2}\mathbf{v}_1 = \frac{1}{2}\mathbf{v}_1 + 0\mathbf{v}_2, \quad \mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2.$$

The weights may be any real numbers, including zero.

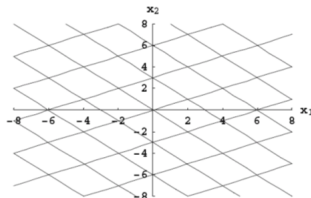
# Example: Linear Combinations in $\mathbb{R}^2$

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}.$$

Express each vector as a linear combination of  $\mathbf{v}_1$  and  $\mathbf{v}_2$ :

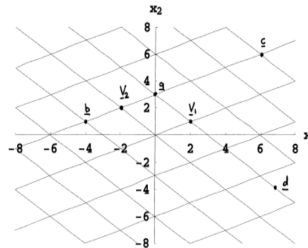
$$\mathbf{a} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} 7 \\ -4 \end{bmatrix}.$$





# Example: Solutions

$$\begin{aligned} \mathbf{a} &= \mathbf{v}_1 + \mathbf{v}_2, & \mathbf{b} &= -\mathbf{v}_1 + \mathbf{v}_2, \\ \mathbf{c} &= 4\mathbf{v}_1 + \mathbf{v}_2, & \mathbf{d} &= \mathbf{v}_1 - \frac{5}{2}\mathbf{v}_2. \end{aligned}$$



## Example: A Vector Equation

Let

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

Determine whether  $\mathbf{b}$  is a linear combination of  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ .

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Determine whether  $\mathbf{b}$  is a linear combination of  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ .

This is equivalent to finding weights  $x_1, x_2, x_3$  such that

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + x_3 \mathbf{a}_3 = \mathbf{b}.$$

# Rewriting the Vector Equation

Use the definitions of scalar multiplication and vector addition to rewrite the vector equation:

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 4 \\ 2 \\ 14 \end{bmatrix} + x_3 \begin{bmatrix} 3 \\ 6 \\ 10 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

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Which is the same as

$$\begin{bmatrix} x_1 \\ 0 \\ 3x_1 \end{bmatrix} + \begin{bmatrix} 4x_2 \\ 2x_2 \\ 14x_2 \end{bmatrix} + \begin{bmatrix} 3x_3 \\ 6x_3 \\ 10x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

# From a Vector Equation to a Linear System

$$\begin{bmatrix} x_1 + 4x_2 + 3x_3 \\ 2x_2 + 6x_3 \\ 3x_1 + 14x_2 + 10x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 8 \\ -5 \end{bmatrix}.$$

By equality of vectors,

$$\begin{cases} x_1 + 4x_2 + 3x_3 = -1, \\ 2x_2 + 6x_3 = 8, \\ 3x_1 + 14x_2 + 10x_3 = -5. \end{cases}$$

# Solving the Vector Equation

The corresponding **augmented matrix**:

$$\left[ \begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right]$$

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The corresponding **augmented matrix**:

$$\left[ \begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array} \right].$$



# Solving the Vector Equation

The corresponding **augmented matrix**:

$$\left[ \begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array} \right].$$

Thus  $x_1 = 1$ ,  $x_2 = -2$ , and  $x_3 = 2$ . Therefore  $\mathbf{b}$  is a linear combination of  $\mathbf{a}_1$ ,  $\mathbf{a}_2$ , and  $\mathbf{a}_3$ .

$$\mathbf{b} = \mathbf{a}_1 - 2\mathbf{a}_2 + 2\mathbf{a}_3.$$

# Vector Equations and Augmented Matrices

A solution of

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{b}$$

is found by solving the linear system whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b} \end{bmatrix}.$$

Review of the last example:  $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$  and  $\mathbf{b}$  are columns of the augmented matrix.

$$\left[ \begin{array}{ccc|c} 1 & 4 & 3 & -1 \\ 0 & 2 & 6 & 8 \\ 3 & 14 & 10 & -5 \end{array} \right].$$

$$\begin{array}{cccc} \uparrow & \uparrow & \uparrow & \uparrow \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b} \end{array}$$

# Vector Equation and Linear System

The vector equation

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b}$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}.$$

In particular,  $\mathbf{b}$  can be generated by a linear combination of  $\mathbf{a}_1, \dots, \mathbf{a}_n$  if and only if the corresponding linear system is consistent.

# Vector Equations and Ordinary Systems

Vector equations  $\iff$  Ordinary systems of equations

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**The key idea of linear algebra** is to find a solution

The solution here is to find all **vectors** satisfying the system...

... that are written as a linear combination of a fixed set  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  of vectors.

# Span $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$

If  $\mathbf{v}_1, \dots, \mathbf{v}_p \in \mathbb{R}^n$ , then the set of all their linear combinations is denoted by

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}.$$

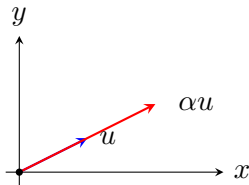
It is the subset of  $\mathbb{R}^n$  **spanned** (or **generated**) by  $\mathbf{v}_1, \dots, \mathbf{v}_p$ ; that is, the collection of all vectors of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p,$$

$c_1, \dots, c_p$  being scalars.

# Span

Span is not "a" vector. It tells us what shapes of vectors can be generated:

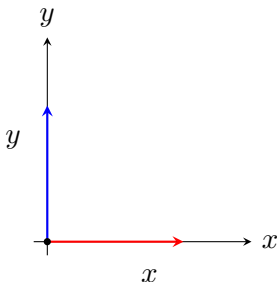


This only has one direction.



# Span

How to use the minimal amount of vectors to span the entire plane?

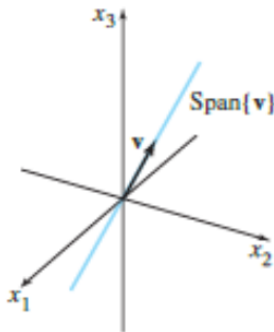


Combining  $\mathbf{x}$  and  $\mathbf{y}$  we can reach any point in the plane.

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\} = \mathbb{R}^2.$$

# A Geometric Description of $\text{Span}\{\mathbf{v}\}$

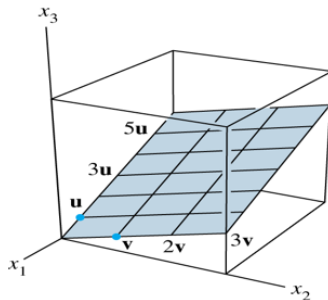
Let  $\mathbf{v}$  be a nonzero vector in  $\mathbb{R}^3$ . Then  $\text{Span}\{\mathbf{v}\}$  is the set of all scalar multiples of  $\mathbf{v}$ , namely the line in  $\mathbb{R}^3$  through  $\mathbf{v}$  and  $\mathbf{0}$ .



**FIGURE 10**  $\text{Span}\{\mathbf{v}\}$  as a line through the origin.

# A Geometric Description of $\text{Span}\{\mathbf{u}, \mathbf{v}\}$

- ▶ If  $\mathbf{u}$  and  $\mathbf{v}$  are nonzero vectors in  $\mathbb{R}^3$ , with  $\mathbf{v}$  **not a scalar multiple** of  $\mathbf{u}$ , then  $\text{Span}\{\mathbf{u}, \mathbf{v}\}$  is the plane in  $\mathbb{R}^3$  containing  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{0}$ .
- ▶ In particular,  $\text{Span}\{\mathbf{u}, \mathbf{v}\}$  contains the line  $\mathbb{R}\mathbf{u}$  through  $\mathbf{u}$  and  $\mathbf{0}$  and the line  $\mathbb{R}\mathbf{v}$  through  $\mathbf{v}$  and  $\mathbf{0}$ .



# Example: Is $\mathbf{b}$ in the Span?

Let

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}.$$

Is  $\mathbf{b}$  in  $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$ ?

$$\left[ \begin{array}{cc|c} 1 & 5 & -3 \\ -2 & -13 & 8 \\ 3 & -3 & 1 \end{array} \right] \sim \left[ \begin{array}{cc|c} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & -18 & 10 \end{array} \right] \sim \left[ \begin{array}{cc|c} 1 & 5 & -3 \\ 0 & -3 & 2 \\ 0 & 0 & -2 \end{array} \right].$$

The third equation is  $0 = -2$ , so the system has no solution. Hence  $\mathbf{b} \notin \text{Span}\{\mathbf{a}_1, \mathbf{a}_2\}$ .

# The Matrix Equation

## 1.4 The Matrix Equation $A\mathbf{x}=\mathbf{b}$

$$A\mathbf{x} = \mathbf{b}$$

# Two Ways to Denote an $m \times n$ Matrix

In terms of the entries of  $A$ :

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}.$$

In terms of its columns:

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n], \quad \mathbf{a}_i \in \mathbb{R}^m \quad (i = 1, \dots, n).$$

# Two Ways to Denote a matrix

In terms of its columns:

$$A = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n], \quad \mathbf{a}_i \in \mathbb{R}^m \quad (i = 1, \dots, n).$$

For example:

$$\mathbf{a}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{i1} \\ \vdots \\ a_{m1} \end{bmatrix}.$$

# Definition of the Matrix–Vector Product

If  $A$  is an  $m \times n$  matrix with columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$ , and  $\mathbf{x} \in \mathbb{R}^n$ , then

$$A\mathbf{x} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Thus  $A\mathbf{x}$  is the **linear combination** of the columns of  $A$  using the corresponding entries of  $\mathbf{x}$  as weights.

$A\mathbf{x}$  is defined only when the number of columns of  $A$  equals the number of entries of  $\mathbf{x}$ . ("Length" of  $A$  = "Height" of  $\mathbf{x}$ )

Moreover,  $\mathbf{a}_i \in \mathbb{R}^m$  and  $A\mathbf{x} \in \mathbb{R}^m$ .



# Example: Computing $Ax$ by Columns

$$\underbrace{\begin{bmatrix} 1 & -4 \\ 3 & 2 \\ 0 & 5 \end{bmatrix}}_A \underbrace{\begin{bmatrix} 7 \\ -6 \end{bmatrix}}_x = 7 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} - 6 \begin{bmatrix} -4 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 7 \\ 21 \\ 0 \end{bmatrix} + \begin{bmatrix} 24 \\ -12 \\ -30 \end{bmatrix} = \underbrace{\begin{bmatrix} 31 \\ 9 \\ -30 \end{bmatrix}}_b.$$

Here,  $A$

## Example: Write a Linear Combination as $A\mathbf{x}$

For  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^m$ , write the linear combination  $3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3$  as a matrix times a vector.

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Answer: place  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$  into the columns of  $A$  and the weights  $3, -5, 7$  into  $\mathbf{x}$ :

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$$3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3 =$$

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For  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^m$ , write the linear combination  $3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3$  as a matrix times a vector.

Answer: place  $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$  into the columns of  $A$  and the weights  $3, -5, 7$  into  $\mathbf{x}$ :

$$3\mathbf{v}_1 - 5\mathbf{v}_2 + 7\mathbf{v}_3 = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} \begin{bmatrix} 3 \\ -5 \\ 7 \end{bmatrix} = A\mathbf{x}.$$

# Three Forms of the Same Linear System

$$\begin{cases} 2x_1 + 3x_2 + 4x_3 = 9, \\ -3x_1 + x_2 = -2. \end{cases}$$

Vector equation:

$$x_1 \begin{bmatrix} 2 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \\ -2 \end{bmatrix}.$$

Matrix equation:

$$\begin{bmatrix} 2 & 3 & 4 \\ -3 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ -2 \end{bmatrix}, \quad A\mathbf{x} = \mathbf{b}.$$

# Three Equivalent Views of a Linear System

A linear system may be viewed

- ① as a system of linear equations;
- ② as a vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b};$$

- ③ as a matrix equation  $A\mathbf{x} = \mathbf{b}$ .

## Theorem 3

If  $A$  is an  $m \times n$  matrix with columns  $\mathbf{a}_1, \dots, \mathbf{a}_n$  and  $\mathbf{b} \in \mathbb{R}^m$ , then the matrix equation

$$A\mathbf{x} = \mathbf{b}$$

has the same solution set as the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b},$$

which has the same solution set as the linear system with augmented matrix

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}.$$

Consequently,  $A\mathbf{x} = \mathbf{b}$  has a solution if and only if  $\mathbf{b}$  is a linear combination of the columns of  $A$ .



# Theorem 4 and Example

For an  $m \times n$  matrix  $A$ , the following are logically equivalent:

- ① For every  $\mathbf{b} \in \mathbb{R}^m$ , the equation  $A\mathbf{x} = \mathbf{b}$  has a solution.
- ② Every  $\mathbf{b} \in \mathbb{R}^m$  is a linear combination of the columns of  $A$ .

# How to determine an equation is consistent for all vectors

EXAMPLE: Let

$$A = \begin{bmatrix} 1 & 4 & 5 \\ -3 & -11 & -14 \\ 2 & 8 & 10 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

Is the equation  $A\mathbf{x} = \mathbf{b}$  consistent for all  $\mathbf{b}$ ?

Answer: augmented matrix corresponding to  $A\mathbf{x} = \mathbf{b}$ :

$$\left[ \begin{array}{ccc|c} 1 & 4 & 5 & b_1 \\ -3 & -11 & -14 & b_2 \\ 2 & 8 & 10 & b_3 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 4 & 5 & b_1 \\ 0 & 1 & 1 & 3b_1 + b_2 \\ 0 & 0 & 0 & -2b_1 + b_3 \end{array} \right].$$

(Can you find a value of  $b_1$  and  $b_3$  that makes  $-2b_1 + b_3$  non zero?)

Hence  $A\mathbf{x} = \mathbf{b}$  is **not** consistent for every  $\mathbf{b}$ .

## Exercise: Consistency for Every $\mathbf{b}$ ?

Compute  $A\mathbf{x}$  where

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Is  $A\mathbf{x} = \mathbf{b}$  consistent for every possible  $\mathbf{b}$ ?

$$\left[ \begin{array}{cc|c} 1 & 2 & b_1 \\ 3 & 4 & b_2 \\ 5 & 6 & b_3 \end{array} \right] \sim \left[ \begin{array}{cc|c} 1 & 0 & -2b_1 + b_2 \\ 0 & 1 & \frac{3b_1 - b_2}{2} \\ 0 & 0 & b_1 - 2b_2 + b_3 \end{array} \right].$$

The equation is not consistent for every  $\mathbf{b}$  because some choices make  $b_1 - 2b_2 + b_3 \neq 0$ .

# Properties of the Matrix–Vector Product $A\mathbf{x}$

## Theorem 5

If  $A$  is an  $m \times n$  matrix,  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ , and  $c$  is a scalar, then

- ①  $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ ;
- ②  $A(c\mathbf{u}) = cA\mathbf{u}$ .

# Computation of $A\mathbf{x}$

Let

$$A = \begin{pmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

From the definition,

$$\begin{aligned} A\mathbf{x} &= x_1 \begin{pmatrix} 2 \\ -1 \\ 6 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} + x_3 \begin{pmatrix} 4 \\ -3 \\ 8 \end{pmatrix} \\ &= \begin{pmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{pmatrix}. \end{aligned}$$

# Computing $A\mathbf{x}$ Row by Row

$$\begin{pmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{pmatrix}.$$

Each output entry is the dot product of one row of  $A$  with  $\mathbf{x}$ :

$$(2, 3, 4)\mathbf{x} = 2x_1 + 3x_2 + 4x_3,$$

$$(-1, 5, -3)\mathbf{x} = -x_1 + 5x_2 - 3x_3,$$

$$(6, -2, 8)\mathbf{x} = 6x_1 - 2x_2 + 8x_3.$$

# Row-Vector Rule for Computing $A\mathbf{x}$

If  $A\mathbf{x}$  is defined, then its  $i$ th entry is the sum of the products of corresponding entries from row  $i$  of  $A$  and from  $\mathbf{x}$ .

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix},$$

$$\begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ s \\ t \end{bmatrix} = \begin{bmatrix} r \\ s \\ t \end{bmatrix}.$$

# Identity Matrix

The  $n \times n$  identity matrix is

$$I = I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

Its diagonal entries are 1 and all other entries are 0. For example,

$$I_3 \begin{bmatrix} r \\ s \\ t \end{bmatrix} = \begin{bmatrix} r \\ s \\ t \end{bmatrix}.$$

In general,

$$I_n \mathbf{x} = \mathbf{x} \quad \text{for every } \mathbf{x} \in \mathbb{R}^n.$$



# Solution Sets of Linear Systems

## 1.5 Solution Sets of Linear Systems

# Topics

- ▶ Parametric Vector Form
- ▶ Solution Sets of Homogeneous Linear Systems
- ▶ Solution Sets of Nonhomogeneous Systems

# Parametric Vector Form

A parametric description

$$\begin{cases} x_1 = 13x_3 - 5x_4, \\ x_2 = 8x_3 - 3x_4, \\ x_3 \text{ is free,} \\ x_4 \text{ is free} \end{cases}$$

becomes, on setting  $x_3 = c_1$  and  $x_4 = c_2$ ,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = c_1 \underbrace{\begin{bmatrix} 13 \\ 8 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{u}} + c_2 \underbrace{\begin{bmatrix} -5 \\ -3 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}}.$$

# Two Descriptions of a Solution Set

## Parametric description

An implicit description of the plane.

## Parametric vector description

An explicit description of the plane as the set spanned by  $\mathbf{u}$  and  $\mathbf{v}$ .

# Solution Sets of Homogeneous Linear Systems

A **homogeneous system** has the form

$$A\mathbf{x} = \mathbf{0},$$

where  $A$  is  $m \times n$  and  $\mathbf{0}$  is the zero vector in  $\mathbb{R}^m$ .

Example:

$$\begin{cases} x_1 + 10x_2 = 0, \\ 2x_1 + 20x_2 = 0, \end{cases} \quad \begin{bmatrix} 1 & 10 \\ 2 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

# Trivial and Nontrivial Solutions

The homogeneous system  $A\mathbf{x} = \mathbf{0}$  always has the **trivial solution** (Trivial: simple, obvious, expected answer).

$$\mathbf{x} = \mathbf{0}.$$

Here  $\mathbf{x}$  and  $\mathbf{0}$  are both vectors. Basically, it is setting all variables to 0. It always works! Nonzero vector solutions are called **nontrivial solutions**.

For the previous example,

$$\left[ \begin{array}{cc|c} 1 & 10 & 0 \\ 2 & 20 & 0 \end{array} \right] \sim \left[ \begin{array}{cc|c} 1 & 10 & 0 \\ 0 & 0 & 0 \end{array} \right].$$

The system is consistent and has a free variable, so it has infinitely many solutions, including nontrivial ones.

## When Do Nontrivial Solutions Exist?

A homogeneous equation  $A\mathbf{x} = \mathbf{0}$  has nontrivial solutions if and only if the system has **at least one free variable**.

Determine whether the following system has nontrivial solutions, and then describe its solution set:

$$\begin{cases} 2x_1 + 4x_2 - 6x_3 = 0, \\ 4x_1 + 8x_2 - 10x_3 = 0. \end{cases}$$

There are three variables but at most two pivots, so at least one variable is free; hence nontrivial solutions exist.

# When Do Nontrivial Solutions Exist?

Why we need to have free variables for nontrivial solution to exist?



# Row Reduction of the Homogeneous System

$$\left[ \begin{array}{ccc|c} 2 & 4 & -6 & 0 \\ 4 & 8 & -10 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 4 & 8 & -10 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right].$$

Therefore

$$x_1 = -2x_2, \quad x_2 \text{ is free}, \quad x_3 = 0.$$

# Parametric Vector Description

Since  $x_1 = -2x_2$ ,  $x_2$  is free, and  $x_3 = 0$ ,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2x_2 \\ x_2 \\ 0 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{v}} = x_2 \mathbf{v}.$$

Thus the solution set is

$$\text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

# Solution Sets of Nonhomogeneous Linear Systems

Describe the solution set of

$$\begin{cases} 2x_1 + 4x_2 - 6x_3 = 0, \\ 4x_1 + 8x_2 - 10x_3 = 4. \end{cases}$$

Nonhomogeneous because (0,4) is not all zero (0,0).

Row reduction gives

$$\left[ \begin{array}{ccc|c} 2 & 4 & -6 & 0 \\ 4 & 8 & -10 & 4 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 1 & 2 & 0 & 6 \\ 0 & 0 & 1 & 2 \end{array} \right].$$

Hence

$$\mathbf{x} = \begin{bmatrix} 6 - 2x_2 \\ x_2 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix}}_{\mathbf{p}} + x_2 \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{v}} = \mathbf{p} + 2\mathbf{v}$$

## Recap of the Previous Two Examples

The solution set of  $A\mathbf{x} = \mathbf{0}$  is

$$\mathbf{x} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = x_2 \mathbf{v}.$$

The solution set of  $A\mathbf{x} = \mathbf{b}$  is

$$\mathbf{x} = \begin{bmatrix} 6 \\ 0 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} = \mathbf{p} + x_2 \mathbf{v}.$$

# Theorem 6

Suppose  $A\mathbf{x} = \mathbf{b}$  is consistent (exists solution) for some given  $\mathbf{b}$  and let  $\mathbf{p}$  be one particular solution. Then its solution set is the set of all vectors of the form

$$\mathbf{w} = \mathbf{p} + \mathbf{v}_h,$$

where  $\mathbf{v}_h$  is any solution of the homogeneous equation  $A\mathbf{x} = \mathbf{0}$ .

**Warning:** This theorem applies only when  $A\mathbf{x} = \mathbf{b}$  has at least one solution. If the  $A\mathbf{x} = \mathbf{b}$  is inconsistent, its solution set is empty.

**CHECK IF SOLUTION EXISTS FIRST!**

## Example: A Homogeneous Plane

Describe the solution set of  $2x_1 - 4x_2 - 4x_3 = 0$  and compare it with the solution set of  $2x_1 - 4x_2 - 4x_3 = 6$ .

## Example: A Homogeneous Plane

Describe the solution set of  $2x_1 - 4x_2 - 4x_3 = 0$  and compare it with the solution set of  $2x_1 - 4x_2 - 4x_3 = 6$ .

Solution: for the first homogeneous equation,

$$\begin{bmatrix} 2 & -4 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & 0 \end{bmatrix}.$$

Therefore

$$\mathbf{x} = \begin{bmatrix} 2x_2 + 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{u}} + x_3 \underbrace{\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}}.$$

## Example: A Parallel Nonhomogeneous Plane

For the second nonhomogeneous equation  $2x_1 - 4x_2 - 4x_3 = 6$ ,

$$\begin{bmatrix} 2 & -4 & -4 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & -2 & 3 \end{bmatrix}.$$

Thus

$$\mathbf{x} = \begin{bmatrix} 3 + 2x_2 + 2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{p}} + x_2 \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{u}} + x_3 \underbrace{\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}_{\mathbf{v}}.$$



## Exercise: True or False?

- ① If  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution, then  $A\mathbf{x} = \mathbf{b}$  has a unique solution.

## Exercise: True or False?

- ① If  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution, then  $A\mathbf{x} = \mathbf{b}$  has a unique solution. **False**  
Has 0 or 1 solution
- ② If  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution, then  $A\mathbf{x} = \mathbf{b}$  has infinitely many solutions.

## Exercise: True or False?

- ① If  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution, then  $A\mathbf{x} = \mathbf{b}$  has a unique solution. **False**  
Has 0 or 1 solution
- ② If  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution, then  $A\mathbf{x} = \mathbf{b}$  has infinitely many solutions. **False**  
Has 0 or infinitely many
- ③ If  $A\mathbf{x} = \mathbf{b}$  has infinitely many solutions, then  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution.

## Exercise: True or False?

- ① If  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution, then  $A\mathbf{x} = \mathbf{b}$  has a unique solution. **False**  
Has 0 or 1 solution
- ② If  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution, then  $A\mathbf{x} = \mathbf{b}$  has infinitely many solutions. **False**  
Has 0 or infinitely many
- ③ If  $A\mathbf{x} = \mathbf{b}$  has infinitely many solutions, then  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution. **True**  
Only trivial

## Exercise: True or False?

● If  $A\mathbf{x} = \mathbf{b}$  has a unique solution, then  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution.

# Exercise: True or False?

- ☐ If  $A\mathbf{x} = \mathbf{b}$  has a unique solution, then  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution. **False**  
 Only nontrivial
- ☐ Any  $n \times n$  system of linear equations has at most  $n$  solutions.

## Exercise: True or False?

- If  $Ax = b$  has a unique solution, then  $Ax = 0$  has a nontrivial solution. **False**  
Only nontrivial
- Any  $n \times n$  system of linear equations has at most  $n$  solutions. **False**  
May be 0, 1, or infinitely many
- The system  $Ax = 0$  has the trivial solution if and only if there is no free variable.

# Exercise: True or False?

- If  $A\mathbf{x} = \mathbf{b}$  has a unique solution, then  $A\mathbf{x} = \mathbf{0}$  has a nontrivial solution. **False**  
Only nontrivial
- Any  $n \times n$  system of linear equations has at most  $n$  solutions. **False**  
May be 0, 1, or infinitely many
- The system  $A\mathbf{x} = \mathbf{0}$  has the trivial solution if and only if there is no free variable. **False**  
Completely reversed. If and only if there is at least one free variable